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BMJ Open

The Value of Robotic Surgery Simulation for Training Surgical Residents and Attendings: A Systematic Review Protocol

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Keywords:	SURGERY, QUALITATIVE RESEARCH, MEDICAL EDUCATION & TRAINING

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Manuscripts

The Value of Robotic Surgery Simulation for Training Surgical Residents and Attendings: A Systematic Review Protocol

Reviewers

Kathryn K. Howard, MD; Hussein Makki; Nathan M Novotny, MD; Misa Mi, PhD MA MLIS; Ngan Nguyen, PhD CHSE

Corresponding Author:
Ngan Nguyen, PhD CHSE
nganngguyen@oakland.edu
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
O'Dowd Hall, Room 454
586 Pioneer Drive
Rochester, MI 48309
United States of America

Authors:
Kathryn K. Howard, MD
William Beaumont Hospital
General Surgery
Royal Oak, MI
United States of America

Hussein Makki, BS
William Beaumont Hospital
Royal Oak, MI
United States of America

Nathan M. Novotny, MD
William Beaumont Hospital
Pediatric Surgery
Royal Oak, MI
United States of America

Misa Mi, PhD MA MLIS
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
Rochester, MI
United States of America

Word Count: 1875 words

Abstract

Introduction

Robotic Surgery is a method of minimally invasive surgery performed through small incisions using a remote robotic console. Surgical residents and attendings participate in simulation training to be able to effectively perform robotic surgery using wet labs, dry labs and virtual reality platforms. Our objective is to identify the effectiveness of robotic simulation on novice robotic surgeons. This review will answer our review question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

Methods and analysis

The PRISMA 2020 statement will be used to guide the conduct of the systematic review. A comprehensive search of PubMed, Embase, the Cochrane Library, Web of Science, Scopus was performed. The studies were then determined to meet initial screening criteria by one individual for abstract and title with full text screening performed by two authors independently and in duplicate. Narrative themes will be collected, analyzed and summarized where possible.

Results

The literature search identified 2089 references which were uploaded into the Covidence software. A total number of 565 studies were removed from screening as duplicates leaving 1524 studies to be screened against inclusion criteria. The title and abstract screening resulted in 119 studies with 1405 studies excluded. Full-text screening is under way.

Ethics and dissemination

There is no IRB approval required given that the work is carried out on previously published papers. The final manuscript and results will be presented and published at an academic conference and peer reviewed journal.

Registration

This protocol is registered with PROSPERO, an international prospective systematic review database. The reviews registration number is: CRD42021274090

Key Words

General Surgery, Education, Simulation and Robotic Surgery

Article Summary- Strengths and Limitations

- A limitation of our review is that it is limited in scope to general surgery training. A large amount of literature on robotic simulation training has been performed by obstetrics and urology.
- A strength of our review is that it is the first comprehensive review to look at outcome data and the value of robotic simulation.
- A strength of our review is the systematic evaluation of the outcome data presented in discrete and tangible levels to organize the prior research.

Article

Introduction:

Robotic Surgery is a method of minimally invasive surgery that utilizes small incisions to perform a variety of surgical procedures. Since 2000, more than 5 million robotic procedures have been

performed with more than 22,000 publications.[1] While the basic principles are similar to laparoscopic surgery in that it results in smaller incisions with less pain, the robot allows for stabilization of instruments in the surgical field and ergonomic comfort for the surgeon.[2] The robot additionally allows for 3D visualization of structures and simultaneous near infrared spectroscopy technology to determine anatomy and blood flow resulting in safer and more effective surgery for the patient.[2] These features of robotic surgery enhance surgeon's skills to care for the patient.

General surgery residents and attendings undergo simulation training in order to gain skills in robotic surgery prior to performing robotic surgery on patients. This allows for safe procedural practice using virtual reality, dry labs using models or wet labs using cadavers or live anesthetized animals. In 2006 SAGES-MIRA met to establish a consensus document on robotic surgery to look at training and credentialing, clinical applications of robots in surgery, risks of surgery and cost-benefit analysis and research. They broke down training into two separate aspects looking at technical training/capability and the training needed for specific operations.[2] At the time of these guidelines, no specific robotic surgical trainer was available but they outlined specific components that they felt were necessary to provide a comprehensive course. They stated that "the course should be taught by instructors with appropriate clinical experience, and should have a curriculum that includes didactic instruction as well as hands-on experience using inanimate and/or animate models. The course director and/or instructor should provide a written assessment of the participant's mastery of course objectives. Documentation for certain courses comprising only didactic instruction may consist of verification of attendance."[2] They additionally state that a course alone is not sufficient to be able to perform a procedure independently and encourage proctoring with observation of live cases being mandatory.

While SAGES recommended a combination of didactic courses, hands-on training and guided operating room components, not all training systems combine these for training and each hospital is allowed to determine their own credentialing and privileges. Due to no standardized training, a variety of different systems have been developed to fill the void, using a combination of didactic only courses, combination courses involving hands on dry or wet labs and/or virtual reality (VR) systems. FRS and DaVinci technology training pathway are didactic only courses designed to set a foundational education system for surgical simulation.[3] Training programs may use these independently or in conjunction with a VR system. There are three main programs that are more comprehensive in their scope with both didactic sessions in addition to in person classes with robotic simulation and cadaver courses. These courses are the SAGES robotics masters series, Robotics Training Network and the Fundamental Skills of Robotic-Assisted Surgery. Trainees may go to these programs exclusively or additionally combine them with other aspects of training. The VR systems work in as an exclusive simulation program with no associated didactic programs but can be combined with didactic sessions.[3]

Our aim is to perform a systematic review of the current literature on robotic skill acquisition of general surgery residents and attendings through simulation and the impact of their learning on the learner, faculty, and patients/hospital. This review will help determine what current literature is available in the field of robotic simulation and guide future research to elucidate the benefit of robotic surgical simulation. The synthesis of information will additionally help programs develop training interventions to teach robotic surgical skills for novice robotic surgeons.

Research Question and Objectives

Question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

The objective is to identify the effectiveness of Robotic Simulation on Novice Robotic Surgeons

More specifically, the objectives are to identify:
The effectiveness of robotic simulation on how it benefits the learner, attendings, patients and hospital system using the Phillips Evaluation Model. The Phillips evaluation method is used to evaluate the benefit of learning by breaking down the costs and benefits to discrete tangible definitions. The Phillips evaluation method has five levels consisting of: Reaction, Learning, Application and Implementation, Impact, and Return on Investment. Specifically in our review we are looking at levels 3, 4 and 5 as it relates to the value of robotic surgery simulation.[4]

Table 1: Levels of Evaluation[4]

Level		Measurement focus
0.	Input	Measures input into the program including the number of participants, costs, and time involvement
1.	Reaction, Satisfaction and Planned Action	Measures participant's reaction to and satisfaction with a training program and participant's plans for action
2.	Learning	Measures increase in knowledge and/or skills, and changes in attitudes
3.	Application and Implementation	Measures transfer of knowledge, skills, and/or attitudes from classroom to the job (change in job behavior due to a training program)
4.	Impact	Measures business and/or healthcare (e.g., patient safety, quality of patient care, etc.) impact
5.	Return on Investment (ROI)	Compares the monetary value of the business and/or healthcare outcomes with the cost of the training program

Methods

The PRISMA 2020 statement will be used to guide the conduct of the systematic review.[5]

Inclusion and Exclusion criteria

Types of participants

The quantitative and qualitative components of this review will include novice robotic general surgery residents and attendings who participate in simulation-based robotic surgery training. These will be people learning robotics for the first time but may have had training in other methods of surgery including open and laparoscopic cases.

Types of intervention(s)/phenomena of interest

The quantitative and qualitative components of the review will consider studies that investigate how novice surgery residents and attendings with no robotic skills acquire their skills through simulation and the value of that simulation using the Phillips Evaluation Model.

Types of outcomes

Studies will be collected to evaluate 5 level outcomes from the use of robotic surgical simulation with the Phillps Evaluation model. Specifically we are looking at application and implementation, impact, and return on investment.[4]

We define Application and Implementation Studies as those that evaluate how the development of robotic skills transfer to the operating room.

We define Impact Studies as those that evaluate how simulation benefits the patient.

We define Return of Investment Studies as those that evaluate the cost of the simulation as it relates to savings in the operating room.

Types of studies

Inclusion criteria:

The quantitative component of the review will consider both experimental and epidemiological study designs including randomized controlled trials, non-randomized controlled trials, quasi-experimental studies, prospective and retrospective cohort studies, case control studies and analytical and descriptive cross-sectional studies for inclusion.

Exclusion criteria

Studies without quantitative or qualitative data will be excluded. Studies looking at robotic simulation in non-general surgeons will be excluded. Studies in foreign languages, without full content, comments, editorials and letters to the editor will be excluded. Studies exclusively looking at level 1 and level 2 outcome data will be excluded.

Patient and Public Involvement

There was no patient or public involvement in the development of the protocol.

Sources of Data

PubMed, Embase, the Cochrane Library, Web of Science, Scopus will be searched for comprehensive literature searches.

Search strategy

The search strategy aimed to find both published and unpublished studies. A three-step search strategy was utilized in this review. An initial limited search of PubMed was undertaken followed by analysis of the text words contained in the title and abstract, and of the index terms used to describe the article. A second search using all identified keywords and index terms was then undertaken across all included databases. Thirdly, a hand search was conducted and the reference lists of all identified reports and articles was searched for additional relevant studies.

The PRISMA - S (Preferred Reporting Items for Systematic reviews and Meta-Analyses literature search extension checklist) will be used to ensure clarity and reproducibility of the search strategies.[6] A combination of index terms and keywords will be used. Examples of search terms include:

(robotic surgical procedures [majr] OR robotics [majr] OR robot*[ti] OR “camera targeting”[ti] OR “PEG transfer”[ti] OR “Da Vinci”[ti] OR “SimSurgery educational platform”[ti] OR “mimic dV-Trainer”[ti] OR “3D systems”[ti]) AND (specialties, surgical/education [mh] OR surgical procedures, operative/education [mh] OR simulation training [mh] OR education [mh] OR education [sh] OR learning [mh] OR clinical competence [mh] OR program evaluation [mh] OR program development[mh] OR train [ti] OR instruct*[ti] OR teach*[ti] OR educat*[ti] OR learn*[ti] OR program*[ti] OR curricul*[ti] OR “clinical skills”[ti] OR “clinical skill”[ti] OR “surgical skills”[ti] OR “surgical skill”[ti] OR “surgical performance”[ti] OR “clinical competency”[ti] OR “clinical competencies”[ti] OR “clinical competence”[ti] OR “program effectiveness”[ti] OR “program evaluation”[ti] OR “program development”[ti] OR ”learning outcome” OR “learning outcomes” OR “training outcome” OR “training outcomes”) AND (Internship and residency [mh] OR fellowships and scholarships [mh] OR education, medical, graduate [mh] OR surgeons [mh:noexp] OR resident* [ti] OR “house staff” [ti] OR residenc*[ti] OR internship*[ti] OR fellows[ti] OR trainee*[ti] OR postgraduat*[ti] OR post-grad*[ti] OR “post graduate”[ti] OR surgeon*[ti])

Assessment of methodological quality

To help prevent duplication the systematic review was registered with PROSPERO which is an international database of prospective systematic reviews. The registration number is: CRD42021274090

Article Screening

Title and abstract screening followed by full-text articles screening will be conducted in Covidence. Title and abstract screening was performed by one individual. Two authors/researchers will conduct full-text screening in duplicates and independently.

Data collection

Quantitative data will be extracted from papers included in the review using the standardized data extraction tool from JBI-MAStARI (Appendix II). The data extracted will include specific details about the interventions, populations, study designs and outcomes of significance to the review question and specific objectives.
Qualitative data will be extracted from papers included in the review using the standardized data extraction tool from JBI-QARI (Appendix I). The data extracted will include specific details about the interventions, populations, study methods and outcomes of significance to the review question and specific objectives.

Data synthesis

Quantitative papers will, where possible, be pooled in statistical meta-analysis. All results will be subject to double data entry. Effect sizes expressed as odds ratio (for categorical data) and weighted mean differences (for continuous data) and their 95% confidence intervals will be calculated for analysis. Heterogeneity will be assessed statistically using the standard I^2 and also explored using subgroup analyses based on the different quantitative study designs included in this review. Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate.
Qualitative research findings will, where possible be pooled using JBI-QARI. This will involve the aggregation or synthesis of findings to generate a set of statements that represent that aggregation, through assembling the findings rated according to their quality, and categorizing these findings on the basis of similarity in meaning. These categories are then subjected to a meta-synthesis and analyzed for themes in order to produce a single comprehensive set of synthesized findings that can be used as a basis for evidence-based practice. Where textual pooling is not possible the findings will be presented in narrative form.

Results

Using the comprehensive search strategy 2089 references were identified and uploaded into the Covidence software. 565 studies were removed from screening as duplications leaving 1524 candidate studies to be screened. 1405 studies did not meet inclusion criteria and were excluded. The remaining 119 studies are to undergo full text screening by 2 independent authors/researchers.

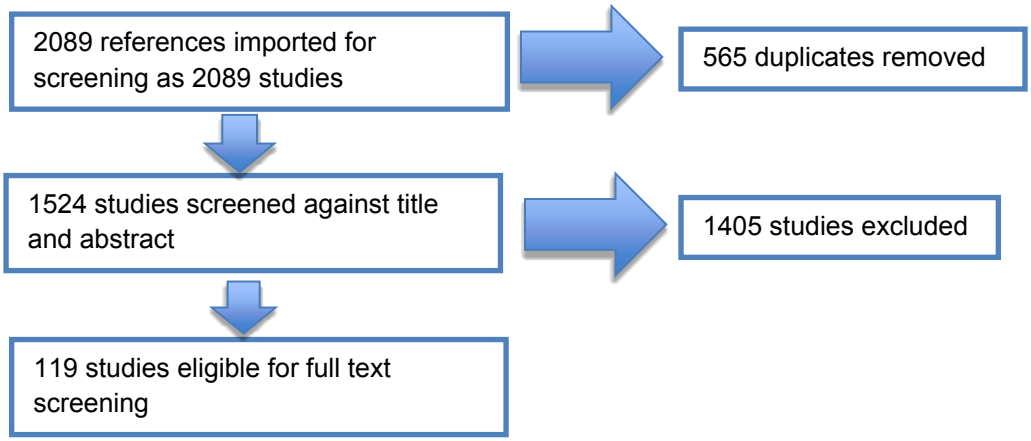


Figure 1: PRISMA depiction of the results obtained through the abstract title screening process

Ethics and Dissemination

No ethical approval from the IRB is required given no direct patient involvement.
Once analysis is completed the results and manuscript will be presented and submitted for publication.

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1. Intuitive.com. https://www.intuitive.com/en-us/healthcare-professionals/surgeons. Published 2021. Accessed August 11, 2021.
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4. Buzachero V, Phillips J, Phillips P, Phillips Z. *Measuring ROI In Healthcare*. New York: McGraw-Hill Education; 2013.
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6. Rethlefsen ML, Kirtley S, Waffenschmidt S, et al. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Syst Rev*. 2021;10(1):39. Published 2021 Jan 26. doi:10.1186/s13643-020-01542-z

Authors' contributions:

Primary authorship by Drs. Kathryn Howard and Misa Mi. Editing performed by Hussein Makki, Drs. Ngan Nguyen and Nathan Novotny.

Funding statement

Pending acceptance to a journal, the article will receive a stipend to cover open access fees through the OUWB library grant.

Competing Interest Statement

We have no competing interests.

Appendix I: QARI data extraction instrument

JBI QARI Data Extraction Form for Interpretive & Critical Research

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Description

Methodology

Method

Phenomena of interest

Setting

Geographical

Cultural

Participants

Data analysis

Authors Conclusions

Comments

Complete Yes ☐ No ☐

Findings	Illustration from Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Extraction of findings complete Yes ☐ No ☐

APPENDIX II: MAStARI data extraction instrument

JBI Data Extraction Form for Experimental / Observational Studies

Reviewer _____ Date _____

Author _____ Year _____

Journal _____ Record Number _____

Study Method

RCT ☐ Quasi-RCT ☐ Longitudinal ☐

Retrospective ☐ Observational ☐ Other ☐

Participants

Setting _____

Population _____

Sample size

Group A _____ Group B _____

Interventions

Intervention A _____

_____

Intervention B _____

_____

Authors Conclusions:

_____

_____

_____

Reviewers Conclusions:

_____

_____

_____

Study results

Dichotomous data

Outcome	Intervention ( ) number / total number	Intervention ( ) number / total number

Continuous data

Outcome	Intervention ( ) number / total number	Intervention ( ) number / total number

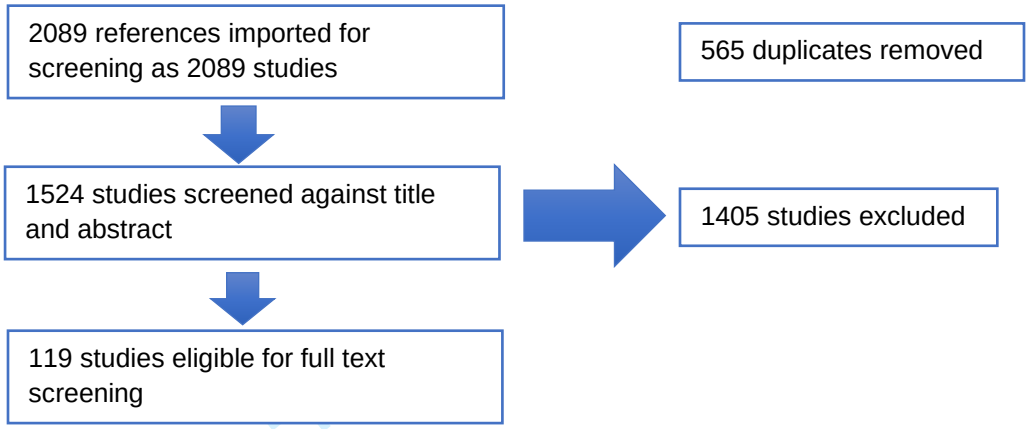


Figure 1: PRISMA depiction of the results obtained through the abstract title screening process

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	2
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	2-3
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	3
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4-5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	5
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	5
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	4-5
Data collection	#9	Specify the methods used to collect data from reports,	6

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process		including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	4
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	4
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	6
Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	6
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study	6

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		intervention characteristics and comparing against the planned groups for each synthesis (item #5))	
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	6
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	6
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	6
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	6
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	6
Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	6
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	6

Results

For peer review only

Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	6
Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	N/A
Study characteristics	#17	Cite each included study and present its characteristics	N/A
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	N/A
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	N/A
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	N/A
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	N/A
Results of syntheses	#20c	Present results of all investigations of possible causes of	N/A

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		heterogeneity among study results	
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	N/A
Risk of reporting biases in syntheses	#21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	N/A
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	N/A
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	N/A
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	N/A
Limitations of the review methods	#23c	Discuss any limitations of the review processes used	N/A
Implications	#23d	Discuss implications of the results for practice, policy, and future research	N/A
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	5
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	N/A

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Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	N/A
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	7
Competing interests	#26	Declare any competing interests of review authors	7
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review	N/A

None The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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BMJ Open

The Value of Robotic Surgery Simulation for Training Surgical Residents and Attendings: A Systematic Review Protocol

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Complete List of Authors:	Howard, Kathryn; William Beaumont Hospital, Surgery Makki, Hussein; William Beaumont Hospital Novotny, Nathan; William Beaumont Hospital, Pediatric Surgery Mi, M; Oakland University William Beaumont School of Medicine, Biomedical Sciences Nguyen, Ngan; Oakland University William Beaumont School of Medicine, Department of Foundational Medical Studies
Primary Subject Heading:	Surgery
Secondary Subject Heading:	Medical education and training
Keywords:	SURGERY, QUALITATIVE RESEARCH, MEDICAL EDUCATION & TRAINING

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Reviewers

Kathryn K. Howard, MD; Hussein Makki; Nathan M Novotny, MD; Misa Mi, PhD MA MLIS; Ngan Nguyen, PhD CHSE

Corresponding Author:
Ngan Nguyen, PhD CHSE
nganngguyen@oakland.edu
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
O'Dowd Hall, Room 454
586 Pioneer Drive
Rochester, MI 48309
United States of America

Authors:
Kathryn K. Howard, MD
William Beaumont Hospital
General Surgery
Royal Oak, MI
United States of America

Hussein Makki, BS
William Beaumont Hospital
Royal Oak, MI
United States of America

Nathan M. Novotny, MD
William Beaumont Hospital
Pediatric Surgery
Royal Oak, MI
United States of America

Misa Mi, PhD MA MLIS
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
Rochester, MI
United States of America

Word Count: 2426 words

Abstract

Introduction

Robotic Surgery is a method of minimally invasive surgery performed through small incisions using a remote robotic console. Surgical residents and attendings participate in simulation training to be able to effectively perform robotic surgery using wet labs, dry labs and virtual reality platforms. Our objective is to identify the effectiveness of robotic simulation on novice robotic surgeons. This review will answer our review question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

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There is no IRB approval required given that the work is carried out on previously published papers. The final manuscript and results will be presented and published at an academic conference and peer reviewed journal.

Registration

This protocol is registered with PROSPERO, an international prospective systematic review database. The reviews registration number is: CRD42021274090

Key Words

General Surgery, Education, Simulation and Robotic Surgery

Article Summary- Strengths and Limitations

- This systematic review organizes abstracts and full text papers using an evidence-based evaluation framework – ROI Methodology.
- This review is limited to novice general surgery residents, fellows and attendings.
- This review uses Covidence software to facilitate cooperation between reviewers.
- This review includes database searches as well as hand searches by a qualified librarian so that all possible publications are included.

Article

Introduction:

Robotic Surgery is a method of minimally invasive surgery that utilizes small incisions to perform a variety of surgical procedures. Since 2000, more than 5 million robotic procedures have been performed with more than 22,000 publications.[1] While the basic principles are similar to laparoscopic surgery in that it results in smaller incisions with less pain, the robot allows for stabilization of instruments in the surgical field and ergonomic comfort for the surgeon.[2] The robot additionally allows for 3D visualization of structures and simultaneous near infrared spectroscopy technology to determine anatomy

and blood flow resulting in safer and more effective surgery for the patient.[2] These features of robotic surgery enhance surgeon's skills to care for the patient.

General surgery residents and attendings undergo simulation training in order to gain skills in robotic surgery prior to performing robotic surgery on patients. This allows for safe procedural practice using virtual reality, dry labs using models or wet labs using cadavers or live anesthetized animals. In 2006 SAGES-MIRA met to establish a consensus document on robotic surgery to look at training and credentialing, clinical applications of robots in surgery, risks of surgery and cost-benefit analysis and research. They broke down training into two separate aspects looking at technical training/capability and the training needed for specific operations.[2] At the time of these guidelines, no specific robotic surgical trainer was available but they outlined specific components that they felt were necessary to provide a comprehensive course. They stated that "the course should be taught by instructors with appropriate clinical experience, and should have a curriculum that includes didactic instruction as well as hands-on experience using inanimate and/or animate models. The course director and/or instructor should provide a written assessment of the participant's mastery of course objectives. Documentation for certain courses comprising only didactic instruction may consist of verification of attendance."[2] They additionally state that a course alone is not sufficient to be able to perform a procedure independently and encourage proctoring with observation of live cases being mandatory.

While SAGES recommended a combination of didactic courses, hands-on training and guided operating room components, not all training systems combine these for training and each hospital is allowed to determine their own credentialing and privileges. Due to no standardized training, a variety of different systems have been developed to fill the void, using a combination of didactic only courses, combination courses involving hands on dry or wet labs and/or virtual reality (VR) systems. FRS and DaVinci technology training pathway are didactic only courses designed to set a foundational education system for surgical simulation.[3] Training programs may use these independently or in conjunction with a VR system. There are three main programs that are more comprehensive in their scope with both didactic sessions in addition to in person classes with robotic simulation and cadaver courses. These courses are the SAGES robotics masters series, Robotics Training Network and the Fundamental Skills of Robotic-Assisted Surgery. Trainees may go to these programs exclusively or additionally combine them with other aspects of training. The VR systems work in as an exclusive simulation program with no associated didactic programs but can be combined with didactic sessions.[3]

Our aim is to perform a systematic review of the current literature on robotic skill acquisition of general surgery residents and attendings through simulation and the impact of their learning on the learner, faculty, and patients/hospital. This review will help determine what current literature is available in the field of robotic simulation and guide future research to elucidate the benefit of robotic surgical simulation. The synthesis of information will additionally help programs develop training interventions to teach robotic surgical skills for novice robotic surgeons.

Research Question and Objectives

Question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

The objective is to identify the effectiveness of Robotic Simulation on Novice Robotic Surgeons.

More specifically, the objectives are to identify:
The effectiveness of robotic simulation on how it benefits the learner, attendings, patients and hospital system using an evidence-based evaluation approach called the ROI Methodology. [4] The ROI Methodology is considered one of the most credible and widely used approaches for demonstrating the impact training. It categorizes the benefits of training into five levels consisting of: Reaction, Learning, Application and Implementation, Impact, and Return on Investment. Specifically in our review we are looking at levels 3, 4 and 5 as it relates to the value of robotic surgery simulation listed below in table 1.[4]

Table 1: Levels of Evaluation [4]

Level	Measurement focus

0.	Input	Measures input into the program including the number of participants, costs, and time involvement
1.	Reaction, Satisfaction and Planned Action	Measures participant's reaction to and satisfaction with a training program and participant's plans for action
2.	Learning	Measures increase in knowledge and/or skills, and changes in attitudes
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5.	Return on Investment (ROI)	Compares the monetary value of the business and/or healthcare outcomes with the cost of the training program

Table 1: ROI Methodology levels of evaluation

Methods

Inclusion and Exclusion criteria

Types of participants

The quantitative and qualitative components of this review will include novice robotic general surgery residents and attendings who participate in simulation-based robotic surgery training. These will be people learning robotics for the first time but may have had training in other methods of surgery including open and laparoscopic cases.

Types of intervention(s)/phenomena of interest

The quantitative and qualitative components of the review will consider studies that investigate how novice surgery residents and attendings with no robotic skills acquire their skills through simulation and the value of that simulation using the ROI Methodology.

Types of outcomes

Studies will be collected to evaluate 5 level outcomes from the use of robotic surgical simulation with the ROI Methodology. Specifically, we are looking at application and implementation, impact, and return on investment.[4]

We define Application and Implementation Studies as those that evaluate how the development of robotic skills transfer to the operating room.

We define Impact Studies as those that evaluate how simulation benefits the patient.

We define Return of Investment Studies as those that evaluate the monetary benefits of simulation-based training relative to the cost of training.

Types of studies

Inclusion criteria:

The quantitative component of the review will consider both experimental and epidemiological study designs including randomized controlled trials, non-randomized controlled trials, quasi-experimental studies, prospective and retrospective cohort studies, case control studies and analytical and descriptive cross-sectional studies for inclusion.

Exclusion criteria

Studies were excluded if they did not have quantitative or qualitative data nor focused on robotic simulation in non-general surgeons. Studies in languages other than English, without translation available as this is the only language all our reviewers share in common, and those without full content (ie. abstracts only), comments, editorials and letters to the editor without qualitative or quantitative data will be excluded. Studies exclusively looking at level 1 and level 2 outcome data will be excluded.

Patient and Public Involvement

There was no patient or public involvement in the development of the protocol.

Sources of Data

PubMed via the National Library of Medicine, Embase via embase.com, the Cochrane Library via Wiley, Web of Science via Clarivate were used for comprehensive literature searches.

Search strategy

The search strategy aimed to find both published and unpublished studies. A three-step search strategy was utilized in this review. A preliminary search of PubMed undertaken followed by analysis of the text words contained in the title and abstract, and of the index terms used to describe studies. A comprehensive literature search using all identified keywords and index terms (unique to each database) was then undertaken across all included databases. Finally, a hand search of the first three pages of google scholar was conducted and the reference lists of all included studies was reviewed for additional relevant studies.

The PRISMA - S (Preferred Reporting Items for Systematic reviews and Meta-Analyses literature search extension checklist) will be used to ensure clarity and reproducibility of the search strategies.[5,6] A combination of index terms and keywords were used. The searches were limited to English language. The search terms utilized are listed below:

PubMed

(robotic surgical procedures [majr] OR robotics [majr] OR robot*[ti] OR “camera targeting”[ti] OR “PEG transfer”[ti] OR “Da Vinci”[ti] OR “SimSurgery educational platform”[ti] OR “mimic dV-Trainer”[ti] OR “3D systems”[ti]) AND (specialties, surgical/education [mh] OR surgical procedures, operative/education [mh] OR simulation training [mh] OR education [mh] OR education [sh] OR learning [mh] OR clinical competence [mh] OR program evaluation [mh] OR program development[mh] OR train [ti] OR instruct*[ti] OR teach*[ti] OR educat*[ti] OR learn*[ti] OR program*[ti] OR curricul*[ti] OR “clinical skills”[ti] OR “clinical skill”[ti] OR “surgical skills”[ti] OR “surgical skill”[ti] OR “surgical performance”[ti] OR “clinical competency”[ti] OR “clinical competencies”[ti] OR “clinical competence”[ti] OR “program effectiveness”[ti] OR “program evaluation”[ti] OR “program development”[ti] OR ”learning outcome” OR “learning outcomes” OR “training outcome” OR “training outcomes”) AND (Internship and residency [mh] OR fellowships and scholarships [mh] OR education, medical, graduate [mh] OR surgeons [mh:noexp] OR resident* [ti] OR “house staff” [ti] OR residenc*[ti] OR internship*[ti] OR fellows[ti] OR trainee*[ti] OR postgraduat*[ti] OR post-grad*[ti] OR “post graduate”[ti] OR surgeon*[ti]) OR (robot*[ti] AND (surgical[tiab] OR surger*[tiab] OR surgery[sh]) AND (residents* [ti] OR residenc*[ti] OR internship*[ti] OR fellows[ti] OR trainee*[ti] OR postgraduat*[ti] OR post-grad*[ti] OR “post graduate”[ti])

Embase

(robot* OR 'camera targeting' OR 'PEG transfer' OR 'Da Vinci' OR 'SimSurgery educational platform' OR 'mimic dV-Trainer' OR '3D systems' OR 'RobotiX mentor') AND (surgical OR surgery OR surgeries) AND (train* OR instruct* OR educat* OR learn* OR teach* OR programs OR program OR curricul* OR 'clinical skills' OR 'clinical skill' OR 'surgical skills' OR 'surgical skill' OR 'surgical performance' OR 'clinical competency' OR 'clinical competencies' OR 'clinical competence' OR 'program effectiveness' OR 'program evaluation' OR 'program development' OR 'learning outcome' OR 'learning outcomes' OR 'training outcome' OR 'training outcomes') AND ('graduate medical education' OR resident* OR residenc* OR 'house staff' OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR 'post graduate' OR (novice next/1 surgeon*)) OR (robot* OR 'camera targeting' OR 'PEG transfer' OR 'Da Vinci' OR 'SimSurgery educational platform' OR 'mimic dV-Trainer' OR '3D systems' OR 'RobotiX mentor') AND (surgical OR surger*) AND ('graduate medical education' OR resident* OR residenc* OR 'house staff' OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR 'post graduate' OR (novice next/1 surgeon*) OR ('robotics'/exp/mj AND 'surgery'/exp AND ('clinical competence'/exp OR 'education'/exp OR simulation training/exp OR 'learning'/exp OR 'training'/exp OR 'program evaluation'/exp OR 'program development'/exp) AND ('residency education'/exp OR resident/exp)

Cochrane Library

((robot* OR "camera targeting" OR "PEG transfer" OR "Da Vinci" OR "SimSurgery educational platform" OR "mimic dV-Trainer" OR "3D systems") AND (surgical OR surger*) AND (train* OR instruct* OR teach* OR educat* OR learn* OR program* OR curricul* OR "clinical skills" OR "clinical skill" OR "surgical skills" OR "surgical skill" OR "surgical performance" OR "clinical competency" OR "clinical competencies" OR "clinical competence"OR "program effectiveness" OR "program evaluation" OR "program development" OR "learning outcome" OR "learning outcomes" OR "training outcome" OR "training outcomes") AND (resident* OR residencies OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate")) OR (robot* AND (surgical OR surger*) AND (residents OR residenc* OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate" OR surgeon*)

Web of Science

(robot* OR "camera targeting" OR "PEG transfer" OR "Da Vinci" OR "SimSurgery educational platform" OR "mimic dV-Trainer" OR "3D systems") AND (surgical OR surger*) AND (train* OR instruct* OR teach* OR educat* OR learn* OR program* OR curricul* OR "clinical skills" OR "clinical skill" OR "surgical skills" OR "surgical skill" OR "surgical performance" OR "clinical competency" OR "clinical competencies" OR "clinical competence"OR "program effectiveness" OR "program evaluation" OR "program development" OR "learning outcome" OR "learning outcomes" OR "training outcome" OR "training outcomes") AND (resident* OR residencies OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate") OR (robot* AND (surgical OR surger*) AND (residents OR residenc* OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate" OR surgeon*)

Assessment of methodological quality

To help prevent duplication the systematic review was registered with PROSPERO which is an international database of prospective systematic reviews. The registration number is: CRD42021274090

Studies will be evaluated for quality using the JBI critical appraisal tools found at jbi.global/critical-appraisal-tools, matching each tool used with its appropriate study type independently and in duplicate.[7]

Article Screening

All search results were imported into Covidence (a web-based software program). Title and abstract screening followed by full-text articles screening will be conducted in Covidence. Title and abstract screening was performed by one individual. Prior to abstract screening by one individual, the first 20 abstracts were analyzed in a meeting with all the reviewers to make sure consensus was achieved on each abstract. Only one author was chosen to review the initial abstracts as the second full text reviewer was not as well versed in surgical robotic curriculum at the time of abstract review. Two authors/researchers will conduct full-text screening in duplicates and independently.

Data collection

Quantitative data will be extracted using an institution created extraction template based on the JBI-MAStARI utilizing the covidence software. The data extracted will include specific details about the interventions, populations, study designs and outcomes of significance to the review question and specific objectives.

Qualitative data will be extracted from papers included in the review using pooled using an institution created extraction template based on the JBI-QARI utilizing Covidence. The data extracted will include specific details about the interventions, populations, study methods and outcomes of significance to the review question and specific objectives.

Data synthesis

Quantitative papers will, where possible, be pooled in statistical meta-analysis. All results will be subject to double data entry. Effect sizes expressed as odds ratio (for categorical data) and weighted mean differences (for continuous data) and their 95% confidence intervals will be calculated for analysis. Heterogeneity will be assessed statistically using the standard I^2 and also explored using subgroup analyses based on the different quantitative study designs included in this review. Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate.

Qualitative research findings will, where possible be pooled using an institution created extraction template based on the JBI-QARI utilizing Covidence. This will involve the aggregation or synthesis of findings to generate a set of statements that represent that aggregation, through assembling the findings rated according to their quality, and categorizing these findings on the basis of similarity in meaning. These categories are then subjected to a meta-synthesis and analyzed for themes in order to produce a single comprehensive set of synthesized findings that can be used as a basis for evidence-based practice. Where textual pooling is not possible the findings will be presented in narrative form.

Deviations

Deviations to the protocol we be published as amendments to the original protocol document.

Ethics and Dissemination

No ethical approval from the IRB is required given no direct patient involvement.

Once analysis is completed the results and manuscript will be presented and submitted for publication.

References

1. Intuitive.com. https://www.intuitive.com/en-us/healthcare-professionals/surgeons. Published 2021. Accessed August 11, 2021.
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6. Rethlefsen ML, Kirtley S, Waffenschmidt S, et al. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Syst Rev*. 2021;10(1):39. Published 2021 Jan 26. doi:10.1186/s13643-020-01542-z
7. *critical-appraisal-tools - Critical Appraisal Tools | Joanna Briggs Institute*. (n.d.). Retrieved February 28, 2022, from https://jbi.global/critical-appraisal-tools

Authors' contributions:

Primary authorship by Drs. Kathryn Howard and Misa Mi. Editing performed by Hussein Makki, Drs. Ngan Nguyen and Nathan Novotny.

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Pending acceptance to a journal, the article will receive a stipend to cover open access fees through the OUWB library grant. Dr. Mi, an author on the protocol, determines eligibility for OUWB grant approval.

Competing Interest Statement

We have no competing interests.

For peer review only

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1, Line 2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2, Line 26-27
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1, Line 9-41
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 8, Line 15-16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Page 7, Line 39-40
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 8, Line 24-25
Sponsor	5b	Provide name for the review funder and/or sponsor	Page 8, Line 24-25
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	Page 8, Line 24-25
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 3, Line 33-36
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 3, Line 39-40, Page 4, Line 6-23
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Page 4, Line 25 – Page 5, Line 2
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 5, Line 13 – Page 6, Line 42
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Page 5, Line 31 – Page 6, Line 42
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 7, Line 5

For peer review only

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) Page 7, Line 5-11
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators Page 7, Line 14-17
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications Page 4, Line 6-23
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale Page 4, Line 16-23
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis Page 7, Line 1-2, Line 32-34
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised Page 7, Line 24-30
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) Page7, Line 24-30
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) Page 7, Line 27-28
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned Page 7, Line 31-37
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) Page 7, Line 1-2
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) Page 7, Line 1-2

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

For peer review only

BMJ Open

The Value of Robotic Surgery Simulation for Training Surgical Residents and Attendings: A Systematic Review Protocol

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Primary Subject Heading:	Surgery
Secondary Subject Heading:	Medical education and training
Keywords:	SURGERY, QUALITATIVE RESEARCH, MEDICAL EDUCATION & TRAINING

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Manuscripts

The Value of Robotic Surgery Simulation for Training Surgical Residents and Attendings: A Systematic Review Protocol

Reviewers

Kathryn K. Howard, MD; Hussein Makki; Nathan M Novotny, MD; Misa Mi, PhD MA MLIS; Ngan Nguyen, PhD CHSE

Corresponding Author:
Ngan Nguyen, PhD CHSE
nganngguyen@oakland.edu
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
O'Dowd Hall, Room 454
586 Pioneer Drive
Rochester, MI 48309
United States of America

Authors:
Kathryn K. Howard, MD
William Beaumont Hospital
General Surgery
Royal Oak, MI
United States of America

Hussein Makki, BS
William Beaumont Hospital
Royal Oak, MI
United States of America

Nathan M. Novotny, MD
William Beaumont Hospital
Pediatric Surgery
Royal Oak, MI
United States of America

Misa Mi, PhD MA MLIS
Oakland University William Beaumont School of Medicine
Department of Foundational Medical Studies
Rochester, MI
United States of America

Word Count: 2426 words

Abstract

Introduction

Robotic Surgery is a method of minimally invasive surgery performed through small incisions using a remote robotic console. Surgical residents and attendings participate in simulation training to be able to effectively perform robotic surgery using wet labs, dry labs and virtual reality platforms. Our objective is to identify the effectiveness of robotic simulation on novice robotic surgeons. This review will answer our review question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

Methods and analysis

. A comprehensive search of PubMed, Embase, the Cochrane Library, and Web of Science was performed. The studies were then determined to meet initial screening criteria by one individual for abstract and title with full text screening performed by two authors independently and in duplicate. Narrative themes will be collected, analyzed and summarized where possible.

Ethics and dissemination

There is no IRB approval required given that the work is carried out on previously published papers. The final manuscript and results will be presented and published at an academic conference and peer reviewed journal.

Registration

This protocol is registered with PROSPERO, an international prospective systematic review database. The reviews registration number is: CRD42021274090

Key Words

General Surgery, Education, Simulation and Robotic Surgery

Article Summary- Strengths and Limitations

- This systematic review organizes abstracts and full text papers using an evidence-based evaluation framework – ROI Methodology.
- This review is limited to novice general surgery residents, fellows and attendings.
- This review uses Covidence software to facilitate cooperation between reviewers.
- This review includes database searches as well as hand searches by a qualified librarian so that all possible publications are included.

Article

Introduction:

Robotic Surgery is a method of minimally invasive surgery that utilizes small incisions to perform a variety of surgical procedures. Since 2000, more than 5 million robotic procedures have been performed with more than 22,000 publications.[1] While the basic principles are similar to laparoscopic surgery in that it results in smaller incisions with less pain, the robot allows for stabilization of instruments in the surgical field and ergonomic comfort for the surgeon.[2] The robot additionally allows for 3D visualization of structures and simultaneous near infrared spectroscopy technology to determine anatomy and blood flow resulting in safer and more effective surgery for the patient.[2] These features of robotic surgery enhance surgeon's skills to care for the patient.

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While SAGES recommended a combination of didactic courses, hands-on training and guided operating room components, not all training systems combine these for training and each hospital is allowed to determine their own credentialing and privileges. Due to no standardized training, a variety of different systems have been developed to fill the void, using a combination of didactic only courses, combination courses involving hands on dry or wet labs and/or virtual reality (VR) systems. FRS and DaVinci technology training pathway are didactic only courses designed to set a foundational education system for surgical simulation.[3] Training programs may use these independently or in conjunction with a VR system. There are three main programs that are more comprehensive in their scope with both didactic sessions in addition to in person classes with robotic simulation and cadaver courses. These courses are the SAGES robotics masters series, Robotics Training Network and the Fundamental Skills of Robotic-Assisted Surgery. Trainees may go to these programs exclusively or additionally combine them with other aspects of training. The VR systems work in as an exclusive simulation program with no associated didactic programs but can be combined with didactic sessions.[3]

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Question: To what extent are robotic simulations for training novice robotic general surgery residents and attendings associated with improved outcomes in comparison with no simulation training?

The objective is to identify the effectiveness of Robotic Simulation on Novice Robotic Surgeons.

More specifically, the objectives are to identify:
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Table 1: Levels of Evaluation [4]

Level	Measurement focus

0.	Input	Measures input into the program including the number of participants, costs, and time involvement
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5.	Return on Investment (ROI)	Compares the monetary value of the business and/or healthcare outcomes with the cost of the training program

Table 1: ROI Methodology levels of evaluation

Methods

Inclusion and Exclusion criteria

Types of participants

The quantitative and qualitative components of this review will include novice robotic general surgery residents and attendings who participate in simulation-based robotic surgery training. These will be people learning robotics for the first time but may have had training in other methods of surgery including open and laparoscopic cases.

Types of intervention(s)/phenomena of interest

The quantitative and qualitative components of the review will consider studies that investigate how novice surgery residents and attendings with no robotic skills acquire their skills through simulation and the value of that simulation using the ROI Methodology.

Types of outcomes

Studies will be collected to evaluate 5 level outcomes from the use of robotic surgical simulation with the ROI Methodology. Specifically, we are looking at application and implementation, impact, and return on investment.[4]

We define Application and Implementation Studies as those that evaluate how the development of robotic skills transfer to the operating room.

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The quantitative component of the review will consider both experimental and epidemiological study designs including randomized controlled trials, non-randomized controlled trials, quasi-experimental studies, prospective and retrospective cohort studies, case control studies and analytical and descriptive cross-sectional studies for inclusion.

Exclusion criteria

Studies were excluded if they did not have quantitative or qualitative data nor focused on robotic simulation in non-general surgeons. Studies in languages other than English, without translation available as this is the only language all our reviewers share in common and no translation services were available, and those without full content (ie. abstracts only), comments, editorials and letters to the editor without qualitative or quantitative data will be excluded. Studies exclusively looking at level 1 and level 2 outcome data will be excluded.

Patient and Public Involvement

There was no patient or public involvement in the development of the protocol.

Sources of Data

PubMed via the National Library of Medicine, Embase via Elsevier at embase.com, the Cochrane Library via Wiley, Web of Science via Clarivate (including Web of Science Core Collection, SciELO Citation Index (2002-present)) were used for comprehensive literature searches.

Search strategy

The search strategy aimed to find both published and unpublished studies. A three-step search strategy was utilized in this review. A preliminary search of PubMed undertaken followed by analysis of the text words contained in the title and abstract, and of the index terms used to describe studies. A comprehensive literature search using all identified keywords and index terms (unique to each database) was then undertaken across all included databases.

The PRISMA - S (Preferred Reporting Items for Systematic reviews and Meta-Analyses literature search extension checklist) will be used to ensure clarity and reproducibility of the search strategies.[5,6] A combination of index terms and keywords were used. The searches were limited to English language. The search terms utilized are listed below:

PubMed

(robotic surgical procedures [majr] OR robotics [majr] OR robot*[ti] OR “camera targeting”[ti] OR “PEG transfer”[ti] OR “Da Vinci”[ti] OR “SimSurgery educational platform”[ti] OR “mimic dV-Trainer”[ti] OR “3D systems”[ti]) AND (specialties, surgical/education [mh] OR surgical procedures, operative/education [mh] OR simulation training [mh] OR education [mh] OR education [sh] OR learning [mh] OR clinical competence [mh] OR program evaluation [mh] OR program development[mh] OR train [ti] OR instruct*[ti] OR teach*[ti] OR educat*[ti] OR learn*[ti] OR program*[ti] OR curricul*[ti] OR “clinical skills”[ti] OR “clinical skill”[ti] OR “surgical skills”[ti] OR “surgical skill”[ti] OR “surgical performance”[ti] OR “clinical competency”[ti] OR “clinical competencies”[ti] OR “clinical competence”[ti] OR “program effectiveness”[ti] OR “program evaluation”[ti] OR “program development”[ti] OR ”learning outcome” OR “learning outcomes” OR “training outcome” OR “training outcomes”) AND (Internship and residency [mh] OR fellowships and scholarships [mh] OR education, medical, graduate [mh] OR surgeons [mh:noexp] OR resident* [ti] OR “house staff” [ti] OR residenc*[ti] OR internship*[ti] OR fellows[ti] OR trainee*[ti] OR postgraduat*[ti] OR post-grad*[ti] OR “post graduate”[ti] OR surgeon*[ti]) OR (robot*[ti] AND (surgical[tiab] OR surger*[tiab] OR surgery[sh]) AND (residents* [ti] OR residenc*[ti] OR internship*[ti] OR fellows[ti] OR trainee*[ti] OR postgraduat*[ti] OR post-grad*[ti] OR “post graduate”[ti])

Embase

(robot* OR 'camera targeting' OR 'PEG transfer' OR 'Da Vinci' OR 'SimSurgery educational platform' OR 'mimic dV-Trainer' OR '3D systems' OR 'RobotiX mentor') AND (surgical OR surgery OR surgeries) AND (train* OR instruct* OR educat* OR learn* OR teach* OR programs OR program OR curricul* OR 'clinical skills' OR 'clinical skill' OR 'surgical skills' OR 'surgical skill' OR 'surgical performance' OR 'clinical competency' OR 'clinical competencies' OR 'clinical competence' OR 'program effectiveness' OR 'program evaluation' OR 'program development' OR 'learning outcome' OR 'learning outcomes' OR 'training outcome' OR 'training outcomes') AND ('graduate medical education' OR resident* OR residenc* OR 'house staff' OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR 'post graduate' OR (novice next/1 surgeon*)) OR (robot* OR 'camera targeting' OR 'PEG transfer' OR 'Da Vinci' OR 'SimSurgery educational platform' OR 'mimic dV-Trainer' OR '3D systems' OR 'RobotiX mentor') AND (surgical OR surger*) AND ('graduate medical education' OR resident* OR residenc* OR 'house staff' OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR 'post graduate' OR (novice next/1 surgeon*) OR ('robotics'/exp/mj AND 'surgery'/exp AND ('clinical competence'/exp OR 'education'/exp OR simulation training/exp OR 'learning'/exp OR 'training'/exp OR 'program evaluation'/exp OR 'program development'/exp) AND ('residency education'/exp OR resident/exp)

Cochrane Library

((robot* OR "camera targeting" OR "PEG transfer" OR "Da Vinci" OR "SimSurgery educational platform" OR "mimic dV-Trainer" OR "3D systems") AND (surgical OR surger*) AND (train* OR instruct* OR teach* OR educat* OR learn* OR program* OR curricul* OR "clinical skills" OR "clinical skill" OR "surgical skills" OR "surgical skill" OR "surgical performance" OR "clinical competency" OR "clinical competencies" OR "clinical competence"OR "program effectiveness" OR "program evaluation" OR "program development" OR "learning outcome" OR "learning outcomes" OR "training outcome" OR "training outcomes") AND (resident* OR residencies OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate")) OR (robot* AND (surgical OR surger*) AND (residents OR residenc* OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate" OR surgeon*)

Web of Science

(robot* OR "camera targeting" OR "PEG transfer" OR "Da Vinci" OR "SimSurgery educational platform" OR "mimic dV-Trainer" OR "3D systems") AND (surgical OR surger*) AND (train* OR instruct* OR teach* OR educat* OR learn* OR program* OR curricul* OR "clinical skills" OR "clinical skill" OR "surgical skills" OR "surgical skill" OR "surgical performance" OR "clinical competency" OR "clinical competencies" OR "clinical competence"OR "program effectiveness" OR "program evaluation" OR "program development" OR "learning outcome" OR "learning outcomes" OR "training outcome" OR "training outcomes") AND (resident* OR residencies OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate") OR (robot* AND (surgical OR surger*) AND (residents OR residenc* OR "house staff" OR internship* OR fellows OR trainee* OR postgraduat* OR post-grad* OR "post graduate" OR surgeon*)

Assessment of methodological quality

To help prevent duplication the systematic review was registered with PROSPERO which is an international database of prospective systematic reviews. The registration number is: CRD42021274090

Studies will be evaluated for quality using the JBI critical appraisal tools found at jbi.global/critical-appraisal-tools, matching each tool used with its appropriate study type independently and in duplicate.[7]

Article Screening

All search results were imported into Covidence (a web-based software program). Title and abstract screening followed by full-text articles screening will be conducted in Covidence. Title and abstract screening was performed by one individual. Prior to abstract screening by one individual, the first 20 abstracts were analyzed in a meeting with all the reviewers to make sure consensus was achieved on each abstract. Only one author was chosen to review the initial abstracts as the second full text reviewer was not as well versed in surgical robotic curriculum at the time of abstract review. Two authors/researchers will conduct full-text screening in duplicates and independently.

Data collection

Quantitative data will be extracted using an institution created extraction template based on the JBI-MAStARI utilizing the covidence software. The data extracted will include specific details about the interventions, populations, study designs and outcomes of significance to the review question and specific objectives.

Qualitative data will be extracted from papers included in the review using pooled using an institution created extraction template based on the JBI-QARI utilizing Covidence. The data extracted will include specific details about the interventions, populations, study methods and outcomes of significance to the review question and specific objectives.

Data synthesis

Quantitative papers will, where possible, be pooled in statistical meta-analysis. All results will be subject to double data entry. Effect sizes expressed as odds ratio (for categorical data) and weighted mean differences (for continuous data) and their 95% confidence intervals will be calculated for analysis. Heterogeneity will be assessed statistically using the standard I^2 and also explored using subgroup analyses based on the different quantitative study designs included in this review. Where statistical pooling is not possible the findings will be presented in narrative form including tables and figures to aid in data presentation where appropriate.

Qualitative research findings will, where possible be pooled using an institution created extraction template based on the JBI-QARI utilizing Covidence. This will involve the aggregation or synthesis of findings to generate a set of statements that represent that aggregation, through assembling the findings rated according to their quality, and categorizing these findings on the basis of similarity in meaning. These categories are then subjected to a meta-synthesis and analyzed for themes in order to produce a single comprehensive set of synthesized findings that can be used as a basis for evidence-based practice. Where textual pooling is not possible the findings will be presented in narrative form.

Deviations

Deviations to the protocol we be published as amendments to the original protocol document.

Ethics and Dissemination

No ethical approval from the IRB is required given no direct patient involvement.

Once analysis is completed the results and manuscript will be presented and submitted for publication.

References

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4. Buzachero V, Phillips J, Phillips P, Phillips Z. *Measuring ROI In Healthcare*. New York: McGraw-Hill Education; 2013.
5. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews
6. Rethlefsen ML, Kirtley S, Waffenschmidt S, et al. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Syst Rev*. 2021;10(1):39. Published 2021 Jan 26. doi:10.1186/s13643-020-01542-z
7. *critical-appraisal-tools - Critical Appraisal Tools | Joanna Briggs Institute*. (n.d.). Retrieved February 28, 2022, from https://jbi.global/critical-appraisal-tools

Authors' contributions:

Protocol idea and planning performed by Drs. Kathryn Howard, Ngan Nguyen and Nathan Novotny. Primary authorship by Drs. Kathryn Howard (introduction, methods) and Misa Mi (methods). Editing performed by Hussein Makki, Drs. Ngan Nguyen and Nathan Novotny. Work submitted by Dr. Kathryn Howard.

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Pending acceptance to a journal, the article will receive a stipend to cover open access fees through the OUWB library grant. Dr. Mi, an author on the protocol, determines eligibility for OUWB grant approval.

Competing Interest Statement

We have no competing interests.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1, Line 2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2, Line 26-27
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1, Line 9-41
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 8, Line 15-16
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Page 7, Line 39-40
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 8, Line 24-25
Sponsor	5b	Provide name for the review funder and/or sponsor	Page 8, Line 24-25
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	Page 8, Line 24-25
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 3, Line 33-36
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 3, Line 39-40, Page 4, Line 6-23
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Page 4, Line 25 – Page 5, Line 2
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 5, Line 13 – Page 6, Line 42
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Page 5, Line 31 – Page 6, Line 42
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 7, Line 5

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) Page 7, Line 5-11
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators Page 7, Line 14-17
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications Page 4, Line 6-23
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale Page 4, Line 16-23
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis Page 7, Line 1-2, Line 32-34
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised Page 7, Line 24-30
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) Page7, Line 24-30
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) Page 7, Line 27-28
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned Page 7, Line 31-37
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) Page 7, Line 1-2
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) Page 7, Line 1-2

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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